

Sukhveer Sohi

SENIOR SOFTWARE ENGINEER

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<https://github.com/IamSohi>

<https://iamsohi.github.io/portfolio/>

Technical Skills:

- **Programming:** Python, JavaScript, TypeScript, SQL, NoSQL, OOP
- **Cloud Computing:** AWS (AWS Cognito, SQS, DynamoDB, Serverless), GCP
- **Frameworks:** Next.js, React, Node.js, Django, Flask, Webpack, OAuth, OKTA, OWASP
- **System Design & Optimization:** Scalable architectures, caching, distributed computing
- **DevOps & CI/CD:** Terraform, Docker, Jenkins, Kubernetes, GitHub Actions
- **Others:** LLMs (Chatbots), MongoDB, PostgreSQL, WebRTC (WebSocket)

Work Experience:

John Deere (Iowa, USA) (Remote)

Aug 2022 – April 2024

Software Engineer (AWS, Python, TypeScript, Java, Node.js, flask)

- **Improved legal team productivity by 70%** by architecting and implementing a scalable microservice with REST APIs, vendor integration, and automated CI/CD pipeline in Python and TypeScript.
- **Optimized processing time by 90% (5h → 30m)** by migrating legacy batch jobs to AWS Step Functions using event-driven and serverless architecture.
- **Enhanced secure communication** across microservices and third-party APIs by **integrating OKTA-issued** access tokens, ensuring authenticated requests and reducing unauthorized access risks.
- **Ensured reliable data delivery** by developing a microservice with retry logic and file validation mechanism to process vendor files and coordinating cross-team debugging to resolve corrupted file issues.
- **Achieved near real-time validation** of large data files (CSV, JSON) within 45 seconds of arrival by deploying scalable AWS Fargate services optimized for high-throughput stream data processing.

Tellex Inc. (Vancouver) (Hybrid)

Nov 2021 – May 2022

Full Stack Web Developer (JavaScript, Next.js (React), Express.js, Serverless)

- **Enabled real-time monitoring of remote security** robots by building responsive **data visualization UIs** using React, Next.js, and Material-UI—enhancing performance and accessibility across devices.
- **Strengthened video streaming security** by using AWS Lambda to generate short-lived STS tokens and enforce OAuth 2.0 policies, protecting real-time transmissions with fine-grained access control.
- Optimized data handling by implementing efficient NoSQL schemas to ensure fast queries and relational-like integrity through cross-team collaboration. Implemented Git Flow for better code management.
- **Cut release cycles by 58%** by leading the adoption of a TDD strategy, introducing clear API contracts with pre/post-condition validation, integrating JEST and Selenium for automated testing across the stack.

ezPT Technologies Ltd. (Surrey) (Remote) Jan 2021 – Oct 2021 (Full-time) Nov 2021 – Nov 2022 (Part-time)

Full Stack Software Engineer (MongoDB, PostgreSQL, React, Django, Material-UI)

- **Secured government funding** by building an MVP full-stack web platform that automated therapist workflows, delivered real-time analytics, and streamlined notetaking and treatment planning.
- **Slashed production time by 96.7%** by implementing a containerized deployment strategy using **Docker and Jenkins** for CI/CD pipelines, while optimizing resource usage with **distributed container orchestration**.
- **Improved team cohesion and onboarding speed** by organizing code reviews, documentation, and technical knowledge-sharing sessions.
- **Enabled rapid delivery and reduced lead time** by establishing Kanban workflows using Monday.com and building a DevOps environment to streamline CI-integrated pipelines.

Personal Projects:

AI-Powered Collaborative Writing Platform

- Enhanced user productivity 30% through real-time, context-aware suggestions by developing a writing tool using LLMs and integrating the OpenAI API for grammar, tone, and style feedback.
- Built with Next.js, TypeScript, Node.js and MongoDB, deploying to AWS to ensure scalability and low latency for multiple simultaneous users. Available at <https://portfolio-projects-writing-platform.vercel.app/>

DevOps - CI/CD with AWS

- Developed and automated a CI/CD pipeline utilizing AWS services including Code Commit, Code Build, Code Deploy, and Code Pipeline.
- Deployed a pre-built website to development and production environments on AWS EC2 servers using the Blue-Green deployment strategy.

Professional Portfolio

- Showcased responsive design and interactivity by implementing and deploying a professional portfolio using React.
- Programmed using JavaScript, HTML5, CSS3, and Material-UI. Accessible at: <https://iamsoshi.github.io/portfolio>

IdoToday- Full Stack Web Application

- Developed a real-time networking website with integrated instant messaging service, achieving SEO optimization with a Page Speed of 3 secs. Avail. at: <https://web.archive.org/web/20171005030146/https://www.idotoday.net/regist/login>
- Utilized the CodeIgniter framework, PHP, SQL, Node.js, Nginx, Apache servers at the back end, & JavaScript, jQuery, HTML5, and Bootstrap at the front-end. Deployed on AWS remote web servers with load balancing for efficient traffic management, coupled with SSL/TLS encryption.

Education:

Kwantlen Polytechnic University, Surrey

May 2018 - May 2020

Computer Science & Information Tech. (4 x Dean's Honor Roll)

Overall GPA: 3.87

Certificates:

Prompt Engineering and LangChain Courses (08/2024 - Present)

Mastered prompt engineering techniques for ChatGPT and developed custom chatbots using LangChain, effectively utilizing diverse data sources to enhance personal and business applications.

Coursera Deep Learning Specialization (03/2019 - 06/2020)

Acquired practical expertise in deep learning, emphasizing Convolutional Networks, RNNs, LSTMs, Adam optimization, Dropout, Batch Normalization, Xavier/He initialization, Natural Language Processing (NLP), and Sequence Models.